$$f_{Dir}(x;\alpha) \propto x_1^{\alpha_1-1} x_2^{\alpha_2-1} \cdots x_D^{\alpha_D-1}$$

$$\frac{\partial f_{Dir}}{\partial x_i \partial x_j} = (\alpha_i-1)(\alpha_j-1)\, x_i^{-1} x_j^{-1} f_{Dir}(x;\alpha)$$

$$\partial x_i \partial x_i = (\alpha_i-1)(\alpha_i-2)\, x_i^{-2} f_{Dir}(x;\alpha)$$

Find the closed form of :

$$\int_\Delta D^{\otimes 2} f_{Dir}(x;\alpha_1) f_{Dir}(x;\alpha_2)\, dx.$$

$$= \frac{C_{Dir}(\alpha_1) C_{Dir}(\alpha_2)}{C_{Dir}(\alpha_1+\alpha_2-1)} \int_\Delta T \cdot f_{Dir}(x;\alpha_1+\alpha_2-1)\, dx.$$

Set $\alpha_1+\alpha_2-1 = \beta$ $\quad$ $\beta^T 1 = \beta_0$

When $i \neq j$ :

$$E_{Dir(\beta)}\left[\frac{1}{x_i x_j}\right] = \frac{(\beta_0-1)(\beta_0-2)}{(\beta_i-1)(\beta_j-1)}$$

$$\frac{C_{Dir}(\alpha_1) C_{Dir}(\alpha_2)}{C_{Dir}(\beta)} \cdot (\alpha_{1i}-1)(\alpha_{1j}-1) \int_\Delta \frac{1}{x_i x_j} f_{Dir}(x;\beta)\, dx$$

$$= \frac{C_{Dir}(\alpha_1) C_{Dir}(\alpha_2)}{C_{Dir}(\beta)} (\beta_0-1)(\beta_0-2) \frac{(\alpha_{1i}-1)(\alpha_{1j}-1)}{(\beta_i-1)(\beta_j-1)}$$

When $i = j$ :

$$E_{Dir(\beta)}\left[\frac{1}{x_i^2}\right] = \frac{(\beta_0-1)(\beta_0-2)}{(\beta_i-1)(\beta_i-2)}$$

$$\frac{C_{Dir}(a_1)\,C_{Dir}(a_2)}{C_{Dir}(\beta)}(a_{1i}-1)(a_{1i}-2)\int_{\Delta}\frac{1}{x_i^2}f_{Dir}(x;\beta)\,dx$$

$$= \frac{C_{Dir}(a_1)\,C_{Dir}(a_2)}{C_{Dir}(\beta)}(\beta_0-1)(\beta_0-2)\frac{(a_{1i}-1)(a_{1i}-2)}{(\beta_i-1)(\beta_i-2)}$$

## First derivative of density on the simplex after ilr transformation.

$x \in \Delta^d$. $\quad y = \text{ilr}(x) \in R^d$, $\quad y \sim N(u, \Sigma)$

Density on $\Delta^d$: $\quad f(x) = \phi_d(y; u, \Sigma)\,|J(x)|$

$$\nabla_x f(x) = f(x)\,\nabla_x \log f(x)$$

$$\begin{aligned}\log f(x) &= \log \phi_d(y; u, \Sigma) + \log|J(x)| \\ &= C - \tfrac{1}{2}(y-u)^T \Sigma^{-1}(y-u) + \log|J(x)|\end{aligned}$$

$$\nabla_y \log \phi_d(y; u, \Sigma) = -\Sigma^{-1}(y-u)$$

$$\nabla_x \log \phi_d(y; u, \Sigma) = (\partial y / \partial x)^T [-\Sigma^{-1}(y-u)]$$

$$\nabla_x \log f(x) = -(\partial \mathrm{ilr}(x)/\partial x)^T \Sigma^{-1}[\mathrm{ilr}(x) - u] + \nabla_x \log |J(x)|$$

$$y = \mathrm{ilr}(x) = V^T \log(x) \qquad \partial y/\partial x = V^T \mathrm{diag}(1/x)$$

$$\nabla_x \log |J(x)| = -[1/x_1, \cdots, 1/x_D] = -x^{\odot(-1)}$$

$$\nabla_x \log f(x) = -\mathrm{diag}(1/x) V \Sigma^{-1}(y-u) - x^{\odot(-1)}$$

$$\nabla_x f(x) = f(x)[-\mathrm{diag}(1/x) V \Sigma^{-1}(y-u) - x^{\odot(-1)}]$$

Proof of : $(r = 1, D)$

$$\int_{\Delta d} D\hat{f}(x;h)^T D f_m(x)\, dx \overset{?}{=} -\mathrm{vec}(I_D)^T \int_{\Delta d} D^{\otimes 2}\hat{f}(x;h)\, f_m(x)\, dx$$

$$\sum_{i=1}^{D} \frac{\partial \hat{f}(x;h)}{\partial x_i} \frac{\partial f_m(x)}{\partial x_i}$$

$$\text{Left hand side} = \sum_{i=1}^{D} \int_{\Delta d} \frac{\partial \hat{f}(x;h)}{\partial x_i} \frac{\partial f_m(x)}{\partial x_i}\, dx.$$

Green's first identity :

$$\int_{\Omega} \frac{\partial u}{\partial x_i} \frac{\partial v}{\partial x_i}\, dx = -\int_{\Omega} \frac{\partial^2 u}{\partial x_i^2} v\, dx + \int_{\partial\Omega} v \frac{\partial u}{\partial x_i} n_i\, dS$$

Let $u(x) = \hat{f}(x;h)$, $v(x) = f_m(x)$

$$\int_{\Delta^d} \frac{\partial \hat{f}}{\partial x_i} \frac{\partial f_m}{\partial x_i} dx = -\int_{\Delta^d} \frac{\partial^2 \hat{f}}{\partial x_i^2} f_m dx + \int_{\partial \Delta^d} f_m \frac{\partial \hat{f}}{\partial x_i} n_i dS$$

Simplex boundary

Sum for $i$: $\int_{\Delta^d} D\hat{f}^T D f_m dx$

$$= -\int_{\Delta^d} \left(\sum_{i=1}^{D} \frac{\partial^2 \hat{f}}{\partial x_i^2}\right) f_m dx + \int_{\partial \Delta^d} f_m(x) D\hat{f}(x;h) \cdot n(x) dS$$

$$= \mathrm{vec}(I_D)^T \mathrm{vec}\left(\frac{\partial^2 \hat{f}}{\partial x_i x_j}\right)_{ij} = \mathrm{vec}(I_D)^T D^{\otimes 2} \hat{f}$$

$$= -\mathrm{vec}(I_D)^T \int_{\Delta^d} D^{\otimes 2} \hat{f} f_m dx + \text{boundary term}$$

If boundary = 0, then equation holds true:

$x \to 0$

$x^\alpha$ {
- $\alpha > 0$ : $0$
- $\alpha = 0$ : $1$
- $\alpha < 0$ : divergence
}

$$\int_{\partial \Delta^d} f_m(x)\, D\hat{f}(x;h) \cdot n(x)\, dS$$

$$f_m(x) = \sum_{i=1}^{m} \pi_i \, Dir(x; a_i) \qquad a_i = [a_{i1}, \cdots a_{iD}]$$

Let $a_{min} := \min a_{ij}$ $\qquad f_m(x) \simeq x_i^{\,a_{min}-1}$

$$D\hat{f}(x;h) = D\frac{1}{n}\sum_{i=1}^{n} Dir(x; \underbrace{X_i/h + C}_{B_i})$$

Let $B_{min} := \min B_{ij}$ $\qquad D\hat{f}(x;h) \simeq x_i^{\,B_{min}-2}$

$$f_m(x)\,\nabla \hat{f}(x;h) \;\asymp\; x_i^{\alpha_{min}-1}\, x_i^{\beta_{min}-2} \;=\; x_i^{\alpha_{min}+\beta_{min}-3}$$

If $\alpha_{min} + \beta_{min} > 3$, then boundary = 0

Consider $\tilde{D}$ rather than $D$:

$$\int_{\Delta^d} \tilde{D}\hat{f}(x;h)^T\, \tilde{D} f_m(x)\, dx \;=\; \int_{\Delta^d} (D\hat{f})^T P\, (D f_m)\, dx$$

$$\Sigma_{i,j=1}^{D}\, \partial_i \hat{f}\, P_{ij}\, \partial_j f_m$$

$$= \Sigma_{i,j=1}^{D} P_{ij} \int_{\Delta^d} \partial_i \hat{f}\, \partial_j f_m\, dx$$

$$\int_{\Delta^d} \partial_i \hat{f}\, \partial_j f_m\, dx \;=\; -\int_{\Delta^d} \partial_i \partial_j \hat{f}\, f_m\, dx \;+\; \int_{\partial\Delta^d} f_m\, \partial_i \hat{f}\, n_j\, dS$$

$$\int_{\Delta^d} \tilde{D}\hat{f}(x;h)^T \tilde{D} f_m(x)\,dx = -\sum_{i,j} P_{ij} \int_{\Delta^d} \partial_i \partial_j \hat{f}\, f_m\, dx + \sum_{i,j} P_{ij} \int_{\partial\Delta^d} f_m\, \partial_i \hat{f}\, n_j(x)\, ds$$

$$= -\int_{\Delta^d} \left(\sum_{i,j} P_{ij} \partial_j \partial_i \hat{f}\right) f_m\, dx + \int_{\partial\Delta^d} f_m \left(\sum_{i,j} P_{ij}\, \partial_i \hat{f}\, n_j(x)\right) ds$$

$$\sum_{i,j} P_{ij} \partial_j \partial_i \hat{f} = D^T P D \hat{f} = (D^T P^T)(PD)\hat{f}$$

$$= \tilde{D}^T \tilde{D} \hat{f} = \mathrm{vec}(I_D)^T \tilde{D}^{\otimes 2} \hat{f}$$

$P = P^T$

$$\sum_{i,j} P_{ij}\, \partial_i \hat{f}\, n_j(x) = (D\hat{f})^T P\, n(x)$$

$$= (PD\hat{f})^T n(x) = (\tilde{D}\hat{f})^T n(x) = \tilde{D}\hat{f} \cdot n(x)$$

$$\int_{\Delta^d} \tilde{D}\hat{f}(x;h)^T \tilde{D} f_m(x)\, dx$$

boundary term.

$$= -\mathrm{vec}(I_D)^T \int_{\Delta^d} \tilde{D}^{\otimes 2} \hat{f} f_m\, dx + \int_{\partial\Delta^d} f_m \tilde{D}\hat{f} \cdot n(x)\, dS$$

Same as before, if $\alpha_{\min} + \beta_{\min} > 3$, then boundary $= 0$